SMART DUSTBIN

Arduino-Based Multifunctional Waste Management Prototype



Smart Dustbin - final prototype

PROJECT REPORT

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Professional Portfolio Edition

Executive Summary

The Smart Dustbin is a mobile waste-management prototype developed to improve hygiene, safety, and convenience. It combines touch-free lid operation, Bluetooth-controlled movement, two waste compartments, gas/smoke alerting, and a mopping mechanism in one system.

The prototype uses separate control sections for mobility and automatic lid operation. The mobility section uses an Arduino with an HC-05 Bluetooth module and four geared motors. The lid section uses an Arduino, an HC-SR04 ultrasonic sensor, and a servo motor. The MQ-2 gas/smoke alert is implemented as an independent circuit with a buzzer, switch, and battery supply.

Key Features

- Touch-free automatic lid using an ultrasonic sensor and servo motor.
- Smartphone-controlled movement through Bluetooth.
- Four geared motors for mobile operation.
- Two chambers for biodegradable and non-biodegradable waste.
- MQ-2 gas/smoke detection with audible buzzer alert.
- Mopping mechanism for cleaning around the dustbin.
- Battery-powered embedded prototype.



Final prototype - angled view

1. Introduction

Growing waste generation and hygiene concerns motivate the use of simple automation in everyday waste-management systems. Conventional bins are usually stationary and often require physical contact for opening and relocation. The Smart Dustbin prototype demonstrates how sensors, motors, wireless control, and safety alerting can be combined to make disposal more convenient and hygienic.

1.1 Purpose and Scope

The project focuses on a practical prototype for household or small public-space use. Its scope includes automatic lid opening, mobile control, physical separation of biodegradable and non-biodegradable waste, gas/smoke alerting, and a mopping feature. The project is a prototype and is intended to demonstrate integration of these functions rather than industrial-scale waste processing.

1.2 System Architecture

Subsystem	Main Components	Function
Automatic Lid	Arduino, HC-SR04, Servo	Detects a nearby hand/object and opens/closes the lid.
Mobility	Arduino, HC-05, Motor Shield/Driver, 4 Gear Motors	Receives Bluetooth commands and drives the wheels.
Gas/Smoke Alert	MQ-2, Buzzer, Switch, Battery	Independent alert circuit for smoke/gas detection.
Waste Storage	Two chambers	Separates biodegradable and non-biodegradable waste.
Cleaning	Mopping mechanism	Supports cleaning around the dustbin.

2. Hardware and Software

2.1 Hardware Components

Arduino boards: Used for the mobility and automatic-lid control sections.

HC-05 Bluetooth module: Provides wireless communication with a smartphone for movement commands.

Motor shield / motor driver: Interfaces the Arduino with the geared DC motors.

Four geared DC motors and wheels: Provide movement of the dustbin.

HC-SR04 ultrasonic sensor: Detects a nearby hand/object for touch-free lid operation.

Servo motor: Mechanically opens and closes the lid.

MQ-2 sensor: Detects smoke/gas for the independent warning circuit.

Buzzer and switch: Provide audible alert and manual power/control for the gas-detection circuit.

Lithium-ion cells / battery supply: Power the prototype subsystems.

Relay and mopping mechanism: Used in the cleaning/mopping section of the prototype.

2.2 Software and Development Tools

Arduino IDE: Used to create and upload Arduino sketches (.ino files).

Servo library: Used for servo control in the automatic-lid sketch.

AFMotor library: Used in the Bluetooth movement sketch with the motor shield.

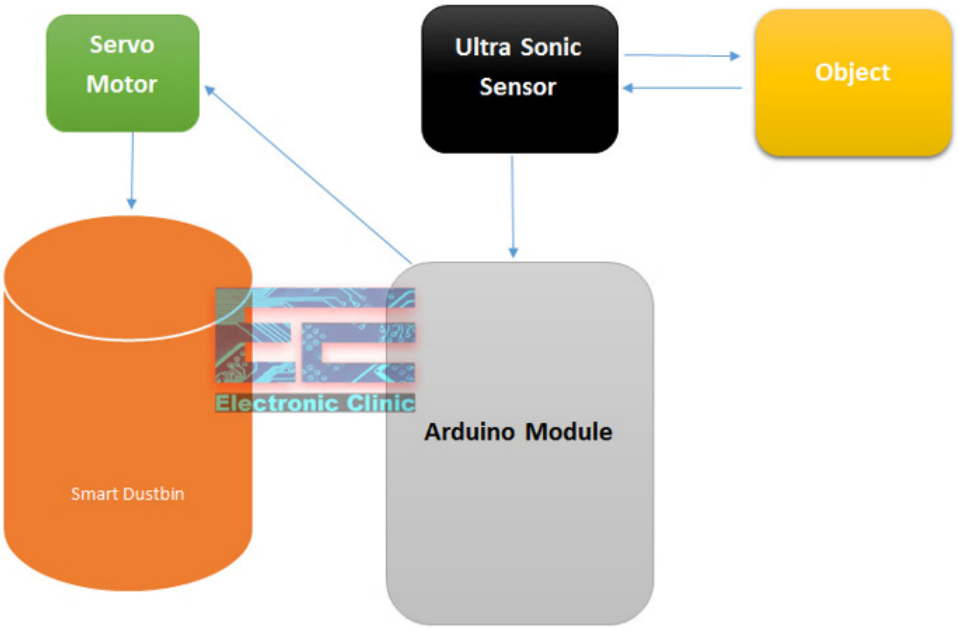
Bluetooth control application: Sends movement commands to the HC-05 module from a smartphone.



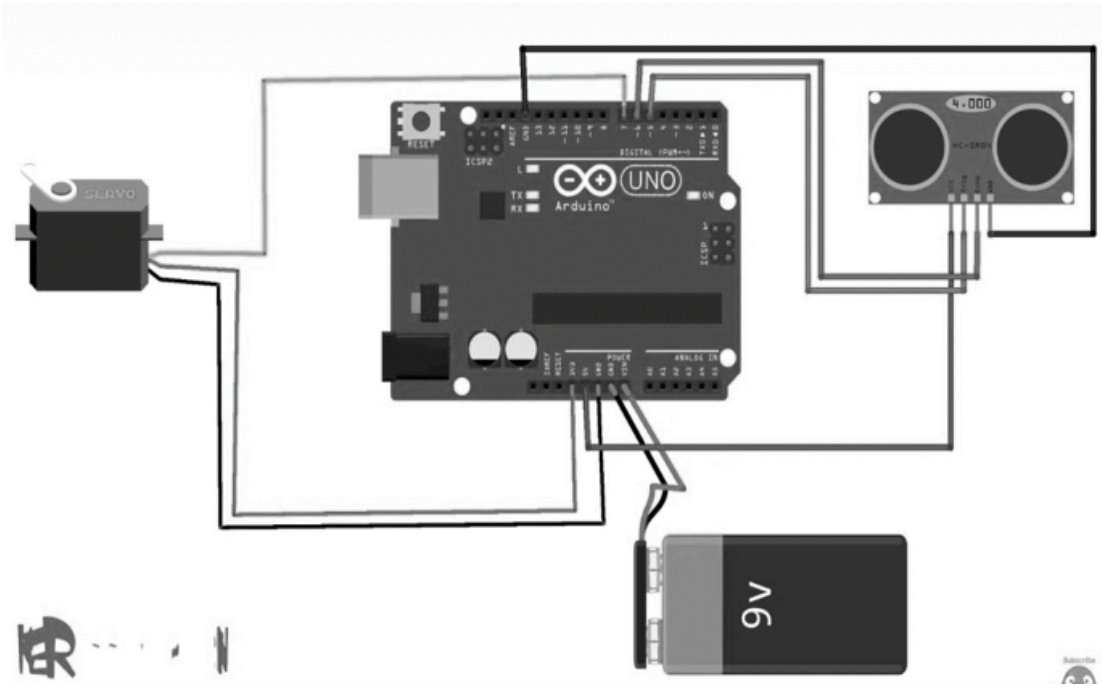
3. Working Principle

3.1 Automatic Lid

The ultrasonic sensor continuously measures the distance to a nearby object. When a hand or object enters the configured detection range, the Arduino commands the servo motor to move the lid to the open position. After a short delay, the servo returns the lid to the closed position.



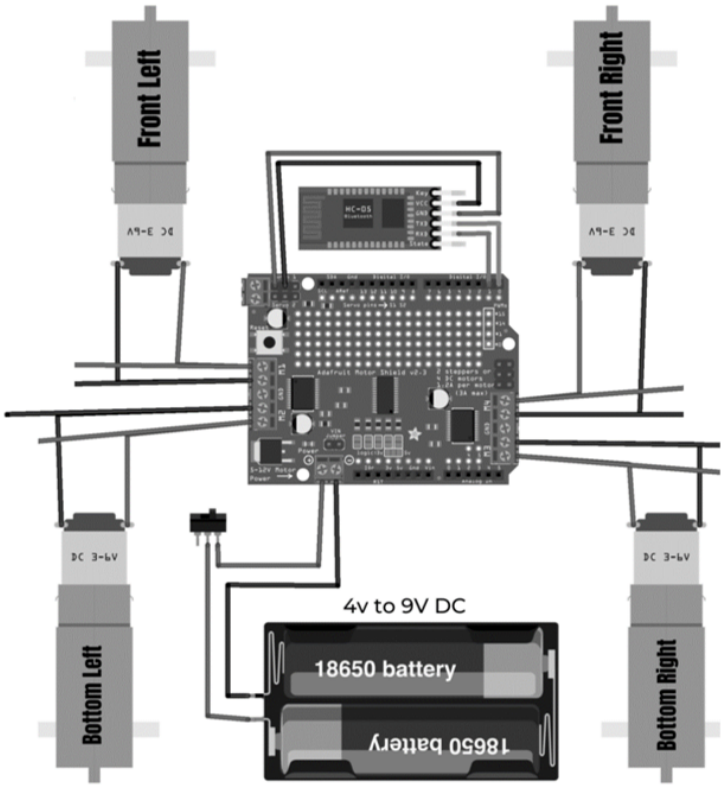
Reference working diagram for the automatic-lid subsystem



Reference circuit diagram: Arduino, HC-SR04 ultrasonic sensor, and servo motor

3.2 Bluetooth-Controlled Mobility

The HC-05 Bluetooth module receives characters sent by a smartphone. The Arduino interprets these commands and controls the four motors through the motor shield/driver. This allows the dustbin to move forward, backward, left, right, stop, and perform additional directional movement supported by the movement sketch.



Reference diagram for Bluetooth-controlled four-motor movement

4. Safety and Waste-Handling Features

4.1 Gas/Smoke Detection

The MQ-2 safety section is independent of the Arduino control logic. The sensor is connected with its own battery supply, switch, and buzzer. When smoke or a detectable gas concentration activates the sensor output, the buzzer provides an audible warning.



Gas/smoke detection testing on the completed prototype

4.2 Dual Waste Compartments

The dustbin contains two physical chambers intended for biodegradable and non-biodegradable waste. This supports source separation of waste. The prototype does not claim automatic material classification; the two compartments provide a practical structure for separated disposal.

4.3 Mopping Mechanism

A mopping mechanism is integrated with the mobile dustbin to demonstrate cleaning around the unit. The feature complements the mobility system and supports the project's hygiene-focused design.

5. Implementation and Testing

The prototype was assembled by integrating the mechanical body, wheels, motors, Arduino control sections, ultrasonic sensor, servo, Bluetooth module, battery supply, gas/smoke alert circuit, and mopping hardware. Individual functions were tested before combined operation.

5.1 Automatic Lid Test

The lid was tested by placing a hand in front of the ultrasonic sensor and verifying that the servo opened the lid and returned it to the closed position after the programmed delay.



Hand approaching the ultrasonic sensor - lid closed



Automatic lid opened after proximity detection

5.2 Mobility Test

Bluetooth commands were sent from a smartphone to verify motor response and movement in the required directions. The four-wheel platform was used to demonstrate that the complete dustbin could be relocated remotely.

5.3 Gas/Smoke Alert Test

The MQ-2 circuit was exposed to a smoke/gas source during testing to verify that the buzzer warning activated as expected.

6. Arduino Sketches

The project repository contains separate Arduino sketches for the two Arduino-controlled subsystems. Keeping them separate reflects the physical architecture of the prototype.

6.1 Automatic Lid - Core Logic

```
#include <Servo.h>

Servo lidServo;

const int trigPin = 9;
const int echoPin = 10;
const int servoPin = 6;
const int detectionDistance = 20;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  lidServo.attach(servoPin);
  lidServo.write(0);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;

  if (distance > 0 && distance <= detectionDistance) {
    lidServo.write(90);
    delay(3000);
    lidServo.write(0);
    delay(1000);
  }
}
```

Note: the repository sketch documents the implemented lid behavior. Pin assignments and servo angles can be adjusted to match the final wiring and mechanical position.

6.2 Bluetooth Movement

The Bluetooth movement sketch uses the HC-05 module, AFMotor library, and four motor objects. Incoming character commands are mapped to forward, backward, left, right, stop, and supported diagonal/directional movements. The complete sketch is kept in the project repository rather than duplicated in this report.

7. Results

The completed prototype demonstrated the main intended functions: touch-free lid opening, mobile operation through Bluetooth, two waste compartments, gas/smoke warning, and the integrated cleaning concept. The real prototype photographs below show the final physical implementation.



Final prototype - front view



Final prototype - side view and mobile wheel base

8. Limitations and Future Improvements

- Improve enclosure finishing and cable management for a more compact production-ready design.
- Add battery-level monitoring and protected charging circuitry.
- Use a dedicated regulated power distribution system for motors, servo, sensors, and control electronics.
- Add bin-level sensing to indicate when either chamber is full.
- Develop a dedicated mobile application with clearer movement and status controls.
- Add automatic waste classification only after integrating and validating suitable sensing hardware.

9. Conclusion

The Smart Dustbin project demonstrates the integration of embedded control, sensing, wireless communication, mobility, and safety alerting in a practical waste-management prototype. The automatic lid reduces the need for direct contact, Bluetooth mobility makes relocation convenient, the two compartments support waste separation, and the independent MQ-2 buzzer circuit adds a useful safety feature. The project also provides a foundation for future improvements such as fill-level monitoring, better power management, and more advanced automation.

10. Project Structure

code/bluetooth_movement/ - Bluetooth movement Arduino sketch
code/automatic_lid/ - Automatic lid Arduino sketch
images/ - Prototype photographs and reference diagrams
docs/ - Project documentation

Source Note

This professional report was rebuilt from the original Smart Dustbin major-project report and the final prototype materials supplied for the project. College/board details, roll number, supervisor/certificate pages, and academic declaration material were intentionally removed for portfolio use.